

Muhammad Furqan Tahir

Software & Automation Engineer

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Portfolio: furqan-portfolio2.vercel.app | GitHub: github.com/furqan-ops | LinkedIn: linkedin.com/in/furqan-ops

SUMMARY

Software & Automation Engineer building practical multi-agent AI systems, internal tools, and web applications. Extensive experience shipping end-to-end workflows with Python, n8n, Playwright, and LLM orchestration. Currently at Applivity automating recruitment pipelines and web data extraction with an uncompromising focus on production reliability: confidence scoring for data accuracy, defensive error handling, and measurable ROI.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML5/CSS3 **Automation & AI:** Multi-Agent Systems, n8n, Playwright, Selenium, LLM Prompting & Function Calling

Web & Databases: Next.js 15, Streamlit, Supabase (PostgreSQL), React, Node.js, Tailwind CSS, Firebase **Tools & Ops:** Git/GitHub, Docker, Power BI, Google Apps Script, REST/Sheets APIs, ERPNext, Vercel

EXPERIENCE

Data Operations Analyst (Automation & Data) | Applivity, Islamabad

May 2024 – Present

- **Resume Pipeline & OCR Accuracy:** Built an n8n workflow to extract candidate data from Google Drive PDFs to Google Sheets. Integrated Claude 3 Haiku for OCR confidence scoring on ambiguous entries—flags unclear details for manual review instead of guessing. Cut manual resume entry by 75% across 5,000+ applicants while maintaining extraction accuracy >99%.
- **MedRim Portal Automation:** Wrote Playwright scripts in Node.js to automate employee dependent registration on the MedRim portal. Automated form-filling across 200+ employee records; added exponential retry logic and explicit waits for network flakiness.
- **Verified Lead Scraping:** Built Python + Selenium scrapers targeting LinkedIn, Upwork, and talent portals to gather verified contact leads. Implemented duplicate detection, data normalization, and direct ingestion into ERPNext with rotating proxies.
- **Hiring & Expense Dashboards:** Created automated Power BI dashboards tracking hiring throughput (pipeline → offer → hire), departmental spend, and budget variance connected to live Google Sheets data, replacing manual monthly executive reporting.

PROJECTS & TECHNICAL WORK

SentinelFlow: Autonomous Incident Response Fleet

github.com/furqan-ops/sentinelflow

- Autonomous self-healing system for data pipelines. A multi-agent fleet (Watchdog, Diagnostician, Remediator) triages webhook failures, performs root cause analysis (RCA) via Gemini 2.0 Flash, and applies auto-patches without human intervention.
- Built an interactive SaaS ops dashboard in Streamlit displaying MTTR benchmarks, SLA uptime health (99.8%), and real-time agent fleet logs.

Autonomous Multi-Agent Dining Concierge & Voice Assistant

multi-ai-agents-customer-support.vercel.app

- Full-stack AI concierge handling bookings, menus, and dietary needs via real-time text and low-latency voice. Built 7 specialized sub-agents orchestrated via LLM routing with multi-turn memory, typo auto-correction, and Web Speech API karaoke text highlighting.

Multi-Agent Performance & Cost Governance Center

multiple-agent-governance-dashboard.streamlit.app

- Real-time Streamlit dashboard tracking multi-agent cluster latency, token consumption, and cost tracking. Implemented semantic caching with Supabase pgvector (embedding distance <0.01), cutting repeat query costs by 40%+ with runaway query breakers.

Attendance Tracking App (Deployed in Production at Applivity)

applivity-attendance-app.web.app

- Cloud-based attendance system replacing paper timesheets for 50+ active staff. Built dual portals (Employee/Admin) for check-in/out stamps, leave requests, and automated monthly payroll calculation with zero hosting costs.

TECHNICAL DEEP DIVES & ARCHITECTURAL DECISIONS

Resume Parsing Reliability & OCR Confidence Scoring:

n8n's native JSON extraction failed on 15–20% of messy resume PDFs (multi-column layouts, inconsistent headings). Solved by inserting a Claude 3 Haiku step to re-extract with OCR and assign confidence scores (0–100) per field. Applied a strict rule: only auto-write high-confidence (>95%) fields; flag medium-confidence entries for rapid human validation. Result: sustained extraction accuracy >99% while automating 75% of ingestion.

pgvector Semantic Caching for LLM Cost Optimization:

The Governance Dashboard constantly queries agent logs for similar incident patterns. Using Supabase pgvector, we cache identical queries (detected via cosine embedding distance <0.01) for 60 seconds. Result: 40% reduction in token consumption and sub-second response times, accepting a controlled 60-second staleness trade-off for operational telemetry.

Web Scraping Resiliency at Scale:

To counter aggressive anti-scraping and rate-limits on professional portals, engineered retry logic with exponential backoff (starting at 2s, capped at 120s) and rotating proxy pools. Replaced arbitrary sleeps with explicit condition waits on critical DOM nodes to eliminate flaky test runs.

Multi-Agent Routing & Failover Architecture:

The Dining Concierge and SentinelFlow route incoming events using strict function schemas. Each specialized sub-agent is scoped to explicit boundaries (e.g. booking agent only executes reservation APIs, never menus). When an agent encounters an anomaly, the supervisor agent catches the error, dispatches an alert, and executes deterministic rollback.

EDUCATION

Bachelor of Software Engineering
International Islamic University, Islamabad
2019 – 2024

CERTIFICATIONS

- Microsoft Power BI Data Analytics
- n8n Workflow Automation Expert
- Claude 101 AI Fluency & Prompt Engineering
- Google Cloud Generative AI Fundamentals